go-On Faculty & Class Advisor Portal

Profile Edit Feature Restoration, Institutional Data Protection & Verification Report

Project Root: E:\originalam	**Target Device:** Samsung Galaxy S24 (1080x2340)
Scope: Faculty & Class Advisor Profile Screen	**Audit Date:** October 19, 2026
Status: Fully Resolved & Verified	**Regression Status:** Zero Regressions (All Tests Passed)

1. Executive Summary & Root Cause Investigation

In the previous release, the Faculty Profile page displayed Account Details, Session Information, Appearance & Theme, and Account Actions, but had **no visible mechanism to edit the faculty profile details or avatar**. Investigation across the Flutter mobile codebase and the FastAPI backend revealed two distinct root causes:

Root Cause 1 (Mobile Client): In mobile/lib/faculty/screens/faculty_profile_screen.dart, the UserAvatarWidget lacked edit badge flags and tap gesture handlers. Furthermore, neither the hero card, the Account Details card header, nor the Account Actions section contained any button or trigger bound to a profile editing sheet.

Root Cause 2 (Backend APIs): In app/api/faculty.py, while student self-service profile updates existed at /api/v1/student/profile, there were no corresponding faculty self-service endpoints (GET /profile, PATCH /profile, PATCH /avatar) to read or persist faculty contact updates.

2. Architectural Remediation & Implementation Details

A secure, end-to-end profile editing architecture was established across both the backend and Flutter mobile client:

Component	Endpoint / File	Responsibility & Business Logic
Backend API	`GET /api/v1/faculty/profile`	Returns authenticated faculty profile including full name, employee ID, email, contact phone, department, resolved avatar ID, role, and class advisor status.
Backend API	`PATCH /api/v1/faculty/profile`	Validates and updates editable fields (full name >= 2 chars, email format & uniqueness, phone 7-15 digits, approved avatar catalog ID). Strictly locks institutional records (employee ID, department, role, status). Logs UPDATE_FACULTY_PROFILE audit trail.
Backend API	`PATCH /api/v1/faculty/avatar`	Enables fast avatar switching with validation against the 16 approved illustrated character catalog.
Mobile UI	`FacultyEditProfileSheet`	Modern, responsive modal bottom sheet (mobile) and centered dialog (desktop) featuring avatar selector, editable contact form, locked institutional records section, and inline error banner.

Mobile Screen	`FacultyProfileScreen`	Restored 4 distinct entry points for profile editing: 1) Avatar edit badge & tap, 2) Hero card 'Edit Profile' pill button, 3) Account Details header 'Edit' button, 4) Account Actions 'Edit Profile Details' action button.

3. Institutional Data Security & Field Locking

To guarantee institutional integrity, strict permission boundaries were implemented at both client and server tiers:

Category	Fields	Permissions & Protection Mechanism
Self-Service Editable	• Full Name • College Email Address • Contact Phone Number • Illustrated Avatar ID	Editable by faculty member. Validated on client and server (min length, regex format, uniqueness check against all accounts, catalog validation).
Locked Institutional Records	• Employee ID / Username • Department • Role Designation (FACULTY/ADMIN) • Class Advisor Assignment • Account Active Status	**Strictly Read-Only (Admin Only).** Displayed with explicit 'Admin Only' lock badge in the UI. Server silently discards or rejects tampering attempts. Cannot be altered via profile update endpoints.

4. Samsung Galaxy S24 Ergonomics & Responsive UX

The implementation was optimized specifically for the **Samsung Galaxy S24** display profile (1080x2340 FHD+, 412x915 logical pixels, 19.5:9 aspect ratio):

- **On-Screen Keyboard Insets Avoidance:** The modal sheet configures margin: EdgeInsets.only(bottom: MediaQuery.of(context).viewInsets.bottom) and wraps the form inside a SingleChildScrollView with isScrollControlled: true. When the virtual keyboard appears, the active text field and action buttons remain fully visible and scrollable without clipping.
- **Zero RenderFlex Overflows:** The hero status container employs Wrap with responsive spacing instead of fixed rows, preventing edge clipping on compact mobile viewports.
- **Theme Persistence:** Profile edits and view modes render cleanly in both **Warm Light** and **Obsidian Dark** liquid glass themes.

5. Verification & Test Execution Summary

Test Suite	Target Scope	Passed	Result
Widget Integration Tests	mobile/test/faculty_profile_edit_test.dart S24 layout, sheet popup, validation, updates, dark theme	5 / 5	**PASSED**
Backend API Tests	tests/test_faculty_profile.py Auth check, validation, sanitization, avatar update	2 / 2	**PASSED**
Dart Static Analysis	flutter analyze lib/ Static type checking, lint rules, null safety	0 issues	**CLEAN**

Audit Sign-Off & Release Certification:
The Faculty Profile editing feature has been fully restored and certified. Personal details and avatar choices are seamlessly editable while institutional college records remain strictly protected against tampering. All unit, widget, and backend tests pass unconditionally.